

1.1) Derivation of the Sigmoid Function from Logistic Regression

We model the conditional probability $P(y = 1 \mid x)$ via the log-odds (logit) as a linear function of x :

$$\log\left(\frac{p}{1-p}\right) = w^\top x + b,$$

where $p = P(y = 1 \mid x)$, now let $z = w^\top x + b$.

Then the logit equation becomes

$$\log\left(\frac{p}{1-p}\right) = z.$$

Exponentiate both sides:

$$\frac{p}{1-p} = e^z.$$

Solve for (p):

$$p = e^z(1-p)$$

$$p = e^z - e^z p$$

$$p + e^z p = e^z$$

$$p(1 + e^z) = e^z$$

$$p = \frac{e^z}{1 + e^z}$$

Rewrite in the standard sigmoid form:

$$p = \frac{1}{1 + e^{-z}} \equiv \sigma(z).$$

Therefore,

$$P(y = 1 \mid x) = \sigma(w^\top x + b) = \frac{1}{1 + e^{-(w^\top x + b)}}.$$

1.2) Derivation of Binary Cross Entropy from MLE

Assume the target label $y \in \{0, 1\}$ follows a Bernoulli distribution. The predicted probability is $\hat{y} = P(y = 1 \mid x)$

The likelihood for a single example is:

$$P(y | x) = \hat{y}^y (1 - \hat{y})^{1-y}$$

For a dataset of m independent examples, the total likelihood is:

$$L = \prod_{i=1}^m \hat{y}_i^{y_i} (1 - \hat{y}_i)^{1-y_i}$$

- Assumes independence across m observations.
- Multiplies individual probabilities to get joint likelihood.

Taking the negative log-likelihood (NLL):

$$\mathcal{L}_{\text{NLL}} = -\log L = -\sum_{i=1}^m [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

- Maximum Likelihood Estimation maximizes the log-likelihood. Equivalently, we minimize the negative log-likelihood

1.3) Why MSE with Sigmoid Activation Is Problematic

Using Mean Squared Error (MSE):

$$L = \frac{1}{2}(\hat{y} - y)^2, \quad \text{where } \hat{y} = \sigma(z)$$

The derivative of the sigmoid function is:

$$\frac{\partial \sigma(z)}{\partial z} = \sigma(z)(1 - \sigma(z)) = \hat{y}(1 - \hat{y})$$

The gradient of the loss with respect to z is:

$$\begin{aligned} \frac{\partial L}{\partial z} &= (\hat{y} - y) \frac{\partial \sigma(z)}{\partial z} \\ \frac{\partial L}{\partial z} &= (\hat{y} - y) \hat{y}(1 - \hat{y}) \end{aligned}$$

When $z \rightarrow -\infty$, $\hat{y} \rightarrow 0$, therefore:

$$(\hat{y} - y) \hat{y}(1 - \hat{y}) \approx 0$$

Similarly, when $z \rightarrow \infty$, $\hat{y} \rightarrow 1$, therefore:

$$(\hat{y} - y) \hat{y}(1 - \hat{y}) \approx 0$$

Even for large values of z , the gradient $\frac{\partial L}{\partial z}$ is close to zero.

This leads to **vanishing gradients** and **slow convergence**.

